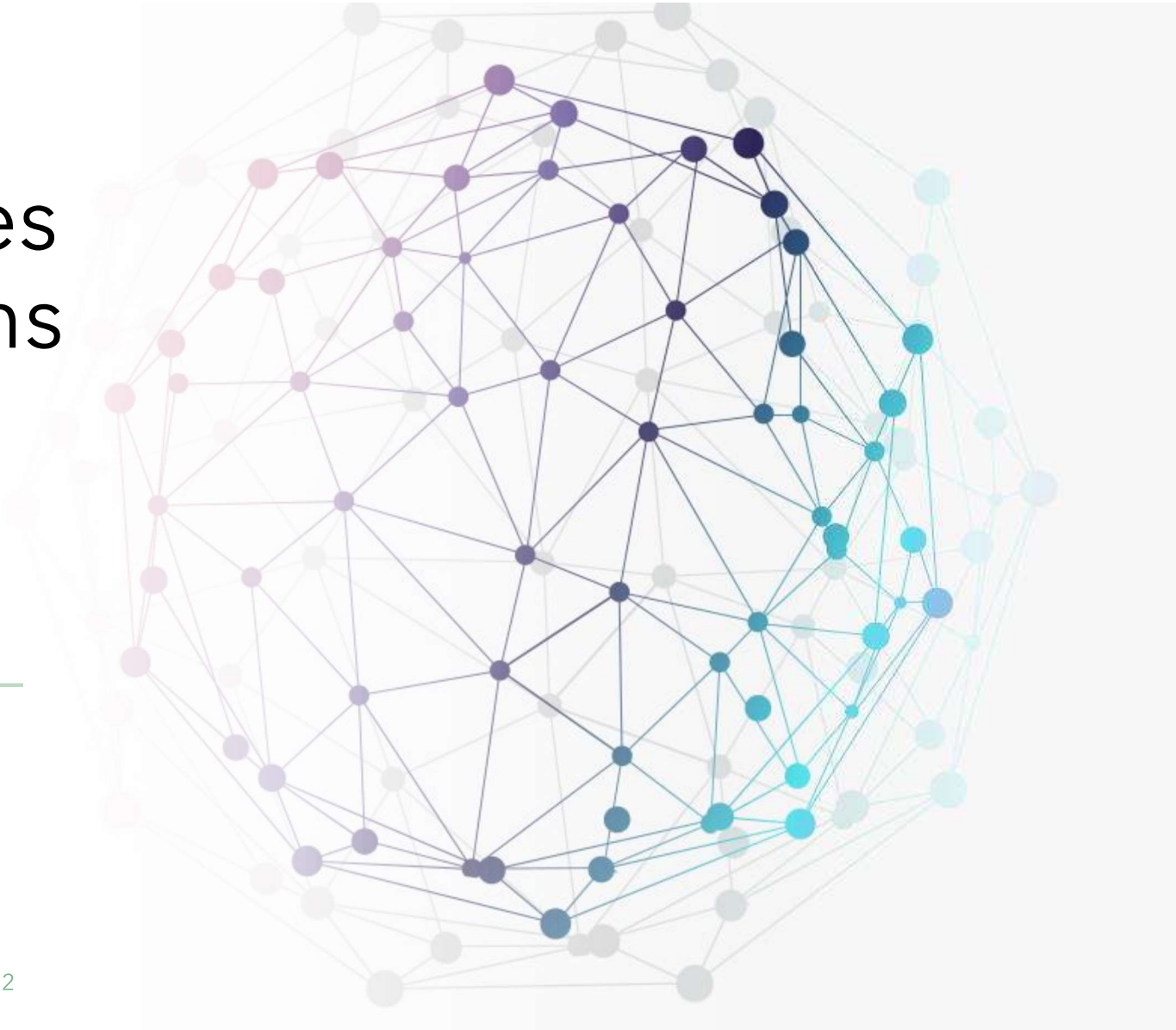
Data Structures and Algorithms

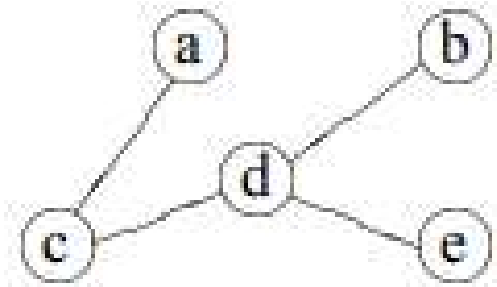


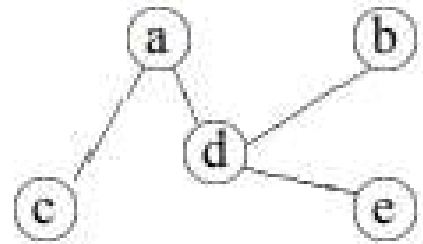
Lecture 8 – Graph Algorithms

Bagwasi 2022

Introduction

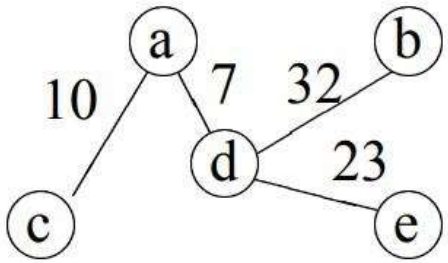
- A Spanning tree has no loops,
- The moment it has loops it now becomes a graph
- Every connected graph has a spanning tree

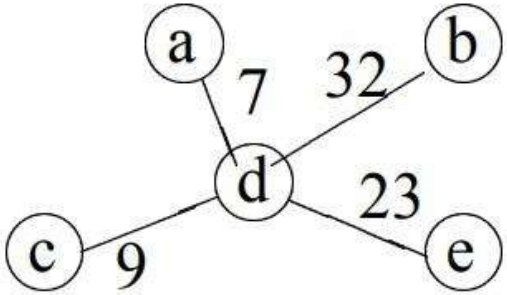




Weighted Graphs

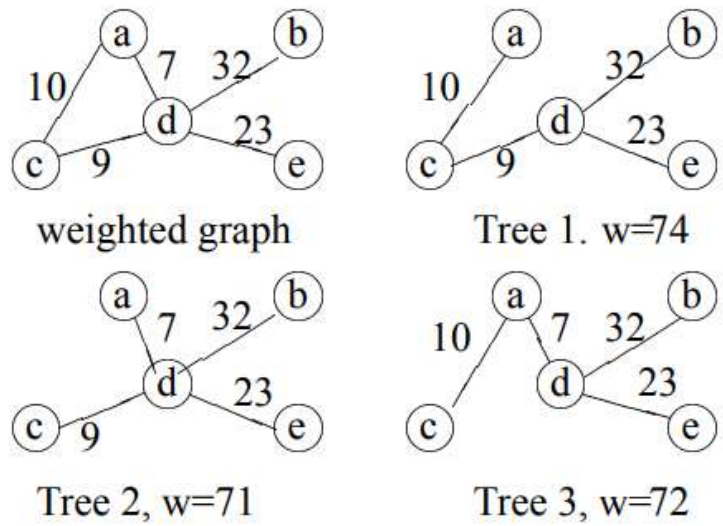
- A weighted graph is a graph in which each edge has a weight (some real number)
- The weight of a graph is the sum of weights of all the edges of a particular graph





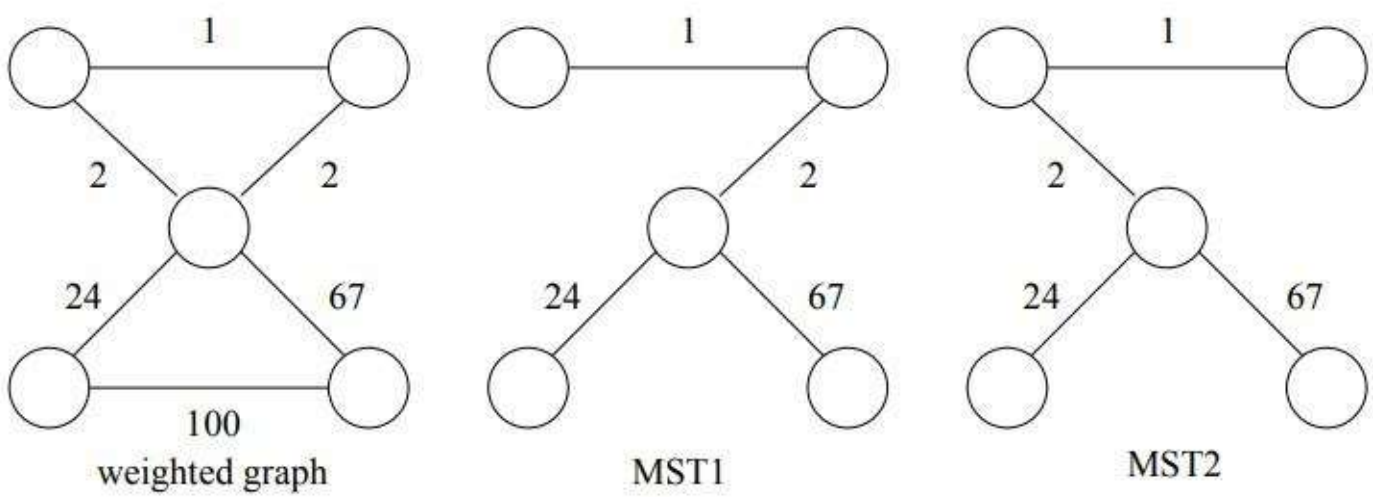
Minimum Spanning Tree

- A minimum spanning tree in an undirected connected weighted graph is a spanning tree of minimum weight



Minimum Spanning Tree

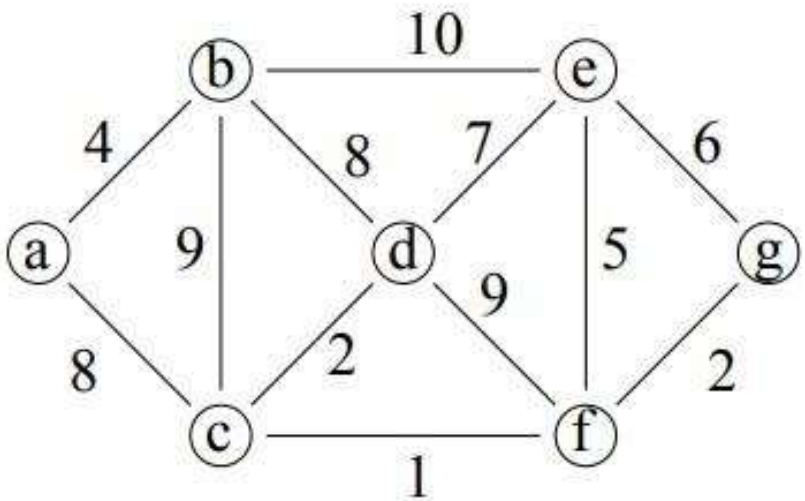
- A Minimum spanning tree may not be unique (We may have different trees with the same cost)



PRIM's ALGORITHM

- Start by picking any vertex r to be root of the tree
- Find the shortest edge leaving the tree and add it to the tree
- Expand the tree by adding the shortest edge leaving from any of the connected edges

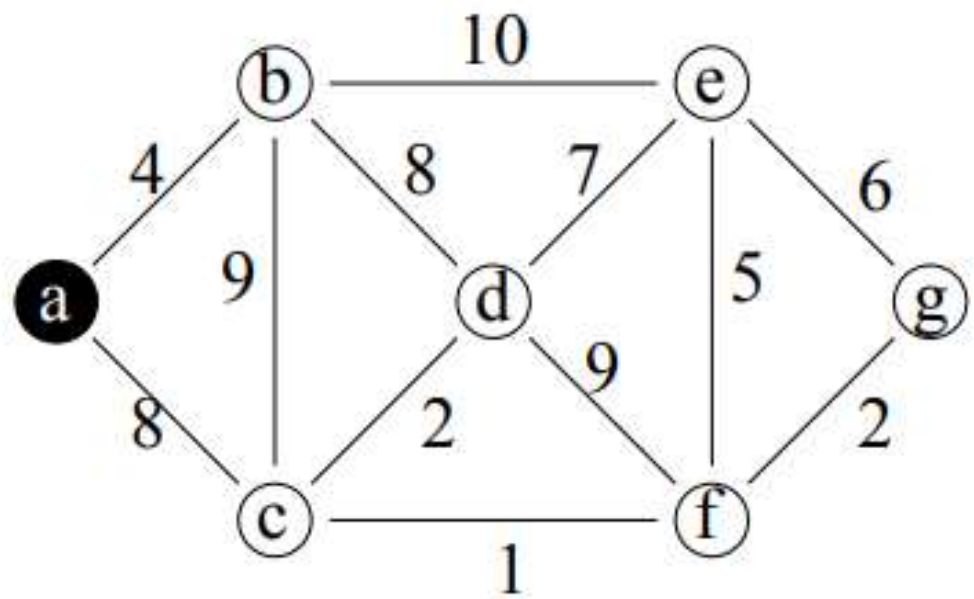
HOW IT WORKS



Connected graph

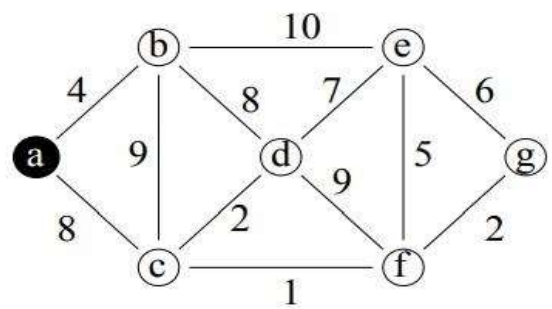
HOW IT WORKS

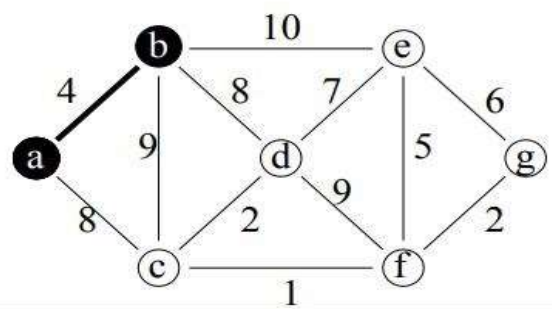
- Pick a starting vertex



HOW IT WORKS

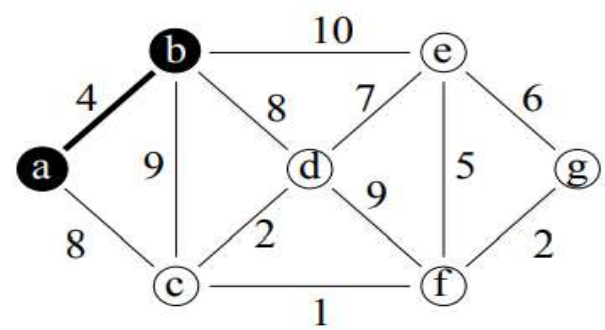
- Now draw a path on the shortest edge of the tree to the next vertex

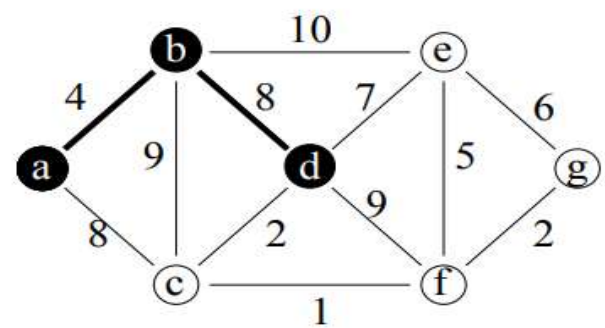




HOW IT WORKS

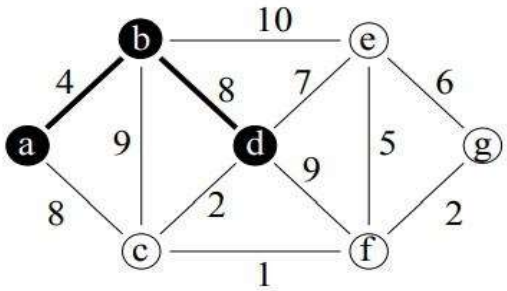
- Now from vertex B, we look for the lowest cost from vertex A and vertex B, the lowest cost in this case is 8 so we draw another connection to either C or D because it's the same cost

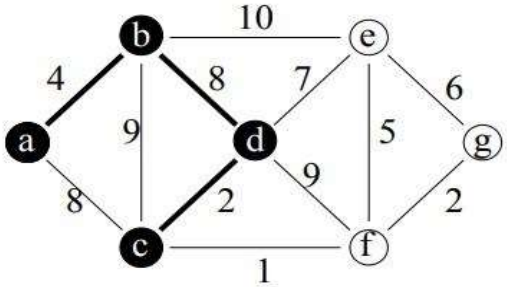




HOW IT WORKS

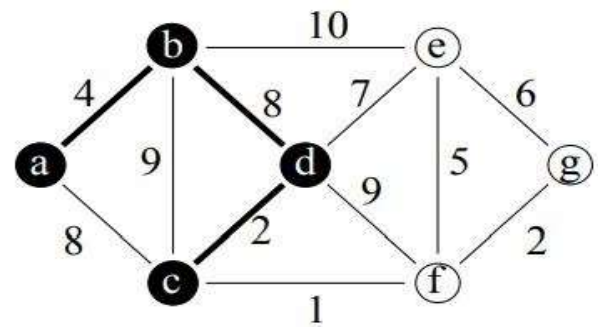
- Now we look for the lowest cost coming from A, B and D and its 2 towards vertex C and we draw it

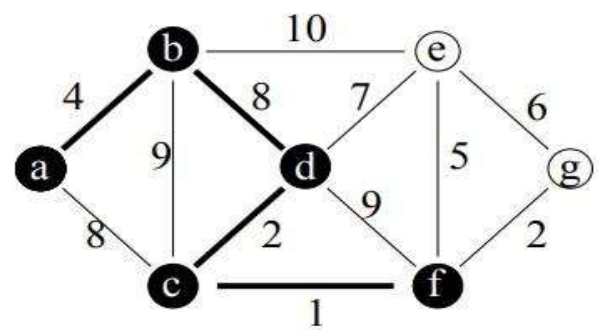




HOW IT WORKS

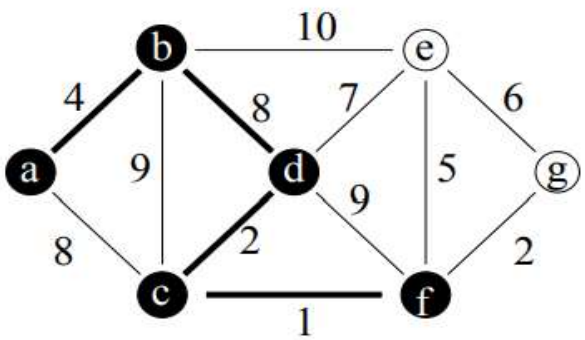
- Now we have visited vertices A,B,D,C
- The next step is to find the shortest distance coming out of any of the vertices
- In this case the cost is 1 towards F then we draw the line

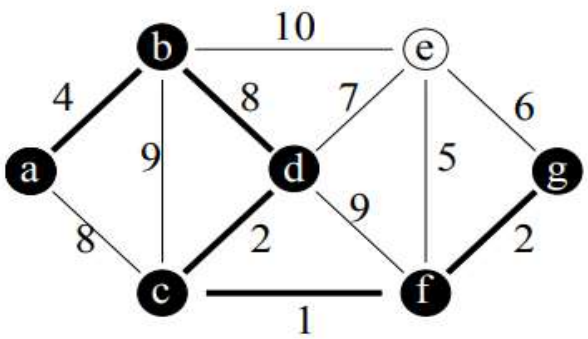




HOW IT WORKS

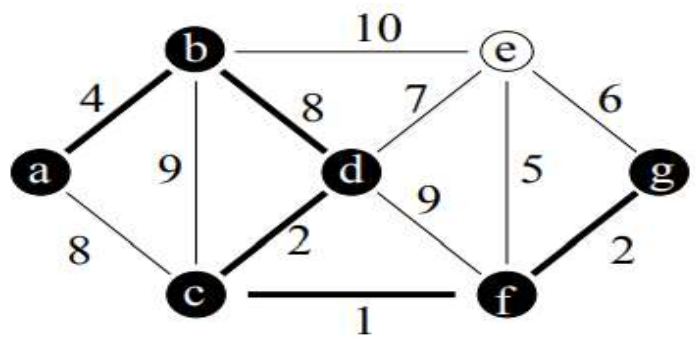
- We now look at the cheapest route out of our connected vertices and we find that its 2 towards G
- We now add the path to our tree

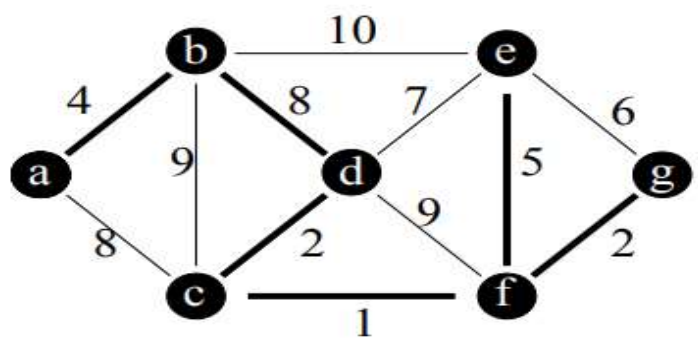




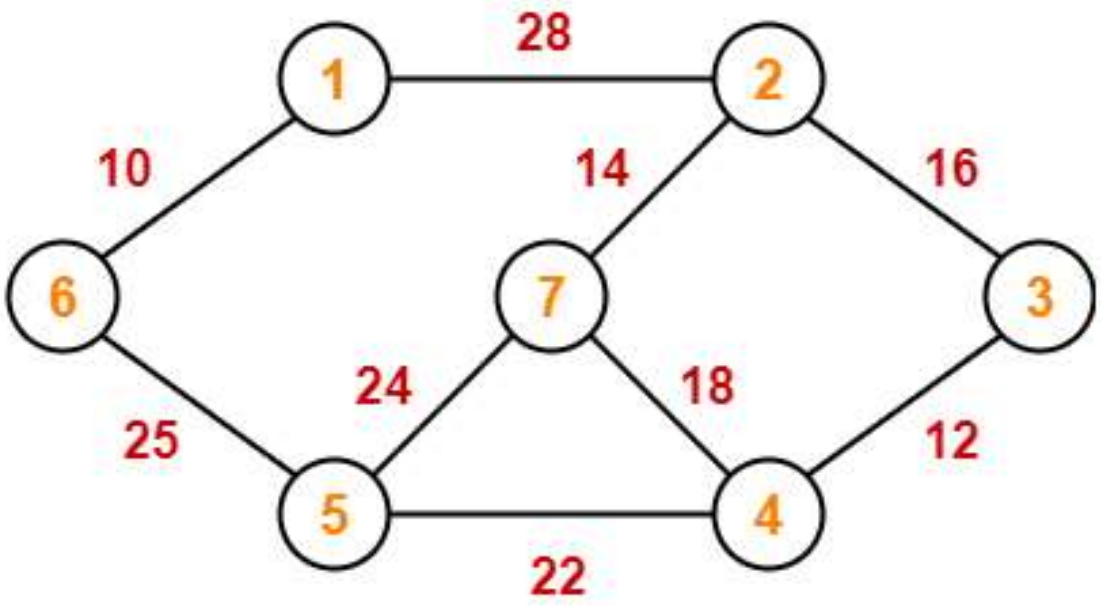
HOW IT WORKS

- Our tree now nears the end
- We find the last shortest cost to connect our final vertex
- and its 5, so we connect it
- The rule is, THERE SHOULD BE NO LOOPS

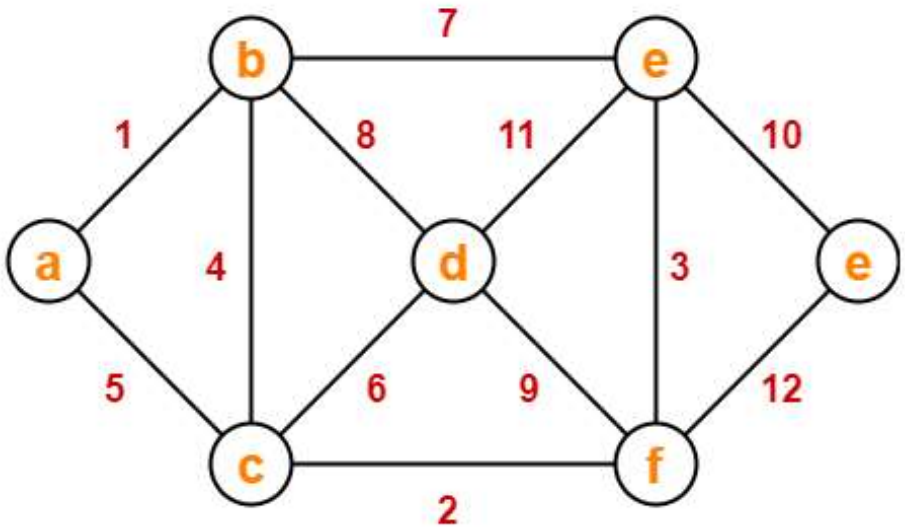




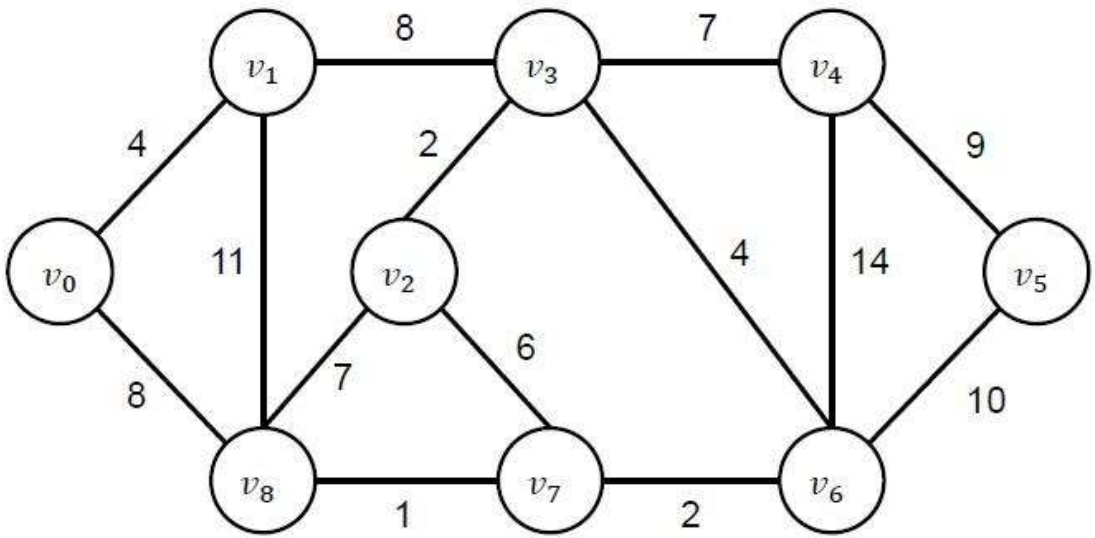
LET'S SOLVE THIS



ANOTHER PROBLEM



ANOTHER PROBLEM

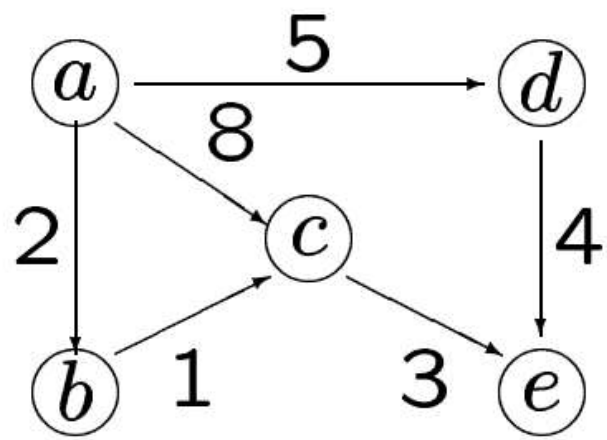


DIJKSTRA'S ALGORITHM

- Given a graph with non negative edge weights, determine the distance and shortest path from the source vertex to every vertex in the graph

SHORTEST PATH

- The distance from a to e is 6

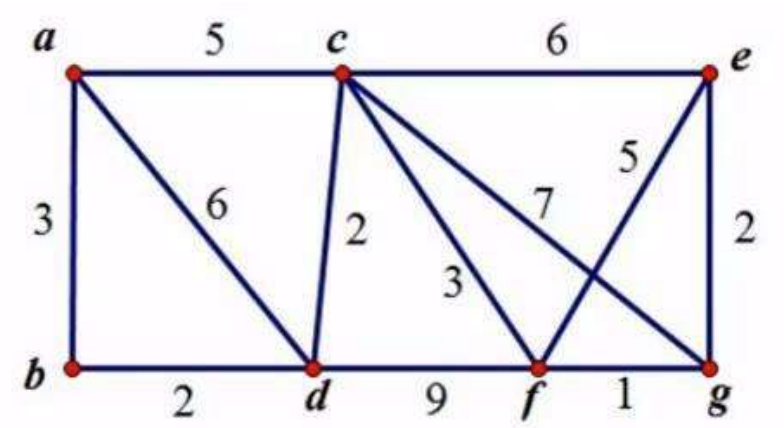


HOW IT WORKS

- Choose the starting vertex
- Calculate the cost to its directly connected neighbours
- Use infinity for the nodes which are not directly connected to the starting vertex
- For the next iteration start with the vertex with the smallest cost

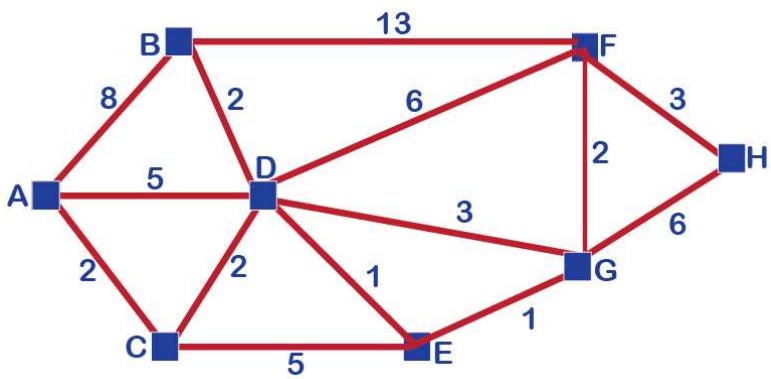
HOW IT WORKS

- Consider the following graph



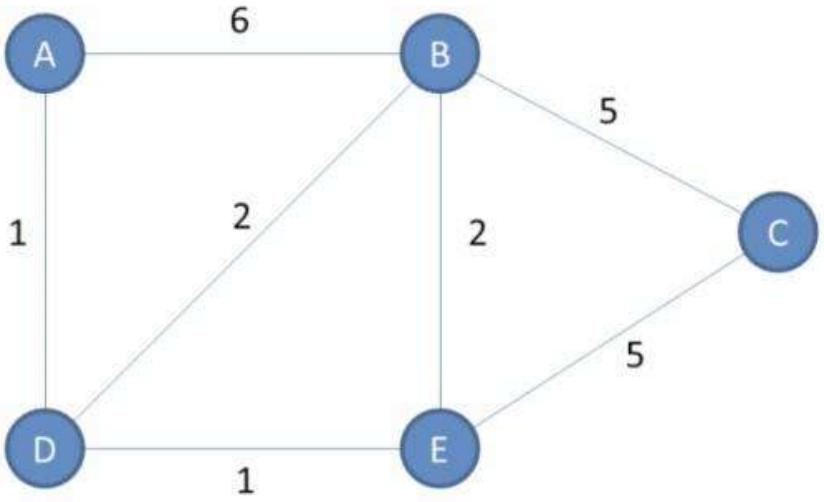
v	a	b	c	d	e	f	g
a	0_a	3_a	5_a	6_a	∞_a	∞_a	∞_a
b	0_a	3_a	5_a	5_b	∞_a	∞_a	∞_a
c	0_a	3_a	5_a	5_b	11_c	8_c	12_c
d	0_a	3_a	5_a	5_b	11_c	8_c	12_c
f	0_a	3_a	5_a	5_b	11_c	8_c	9_f
g	0_a	3_a	5_a	5_b	11_c	8_c	9_f

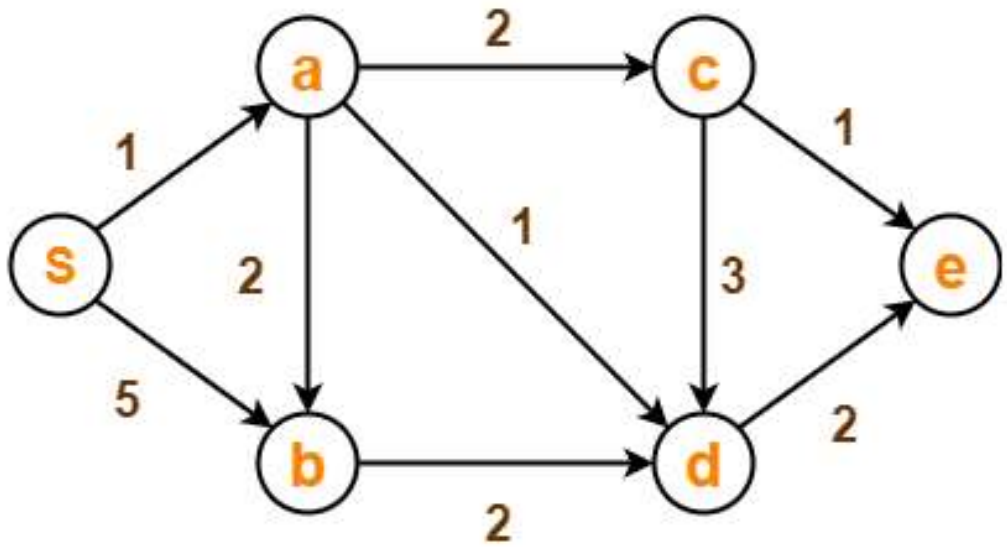
ANOTHER PROBLEM SET



V	A	B	C	D	E	F	G	H
A	0_A	8_A	2_A	5_A	∞	∞	∞	∞
C		8_A	2_A	4_C	7_C	∞	∞	∞
D		6_D		4_C	5_D	10_D	7_D	∞
E		6_D			5_D	10_D	6_E	∞
B		6_D				10_D	6_E	∞
G						8_G	6_E	12_G
F						8_G		11_F
H								11_F

Another Problem



	A	B	C	D	E	
A	0_A	6_A	∞	1_A	∞	
D		3_D	∞	1_A	2_D	
E		3_D	7_D	1_A	2_D	
B			7_D			

TAKE HOME